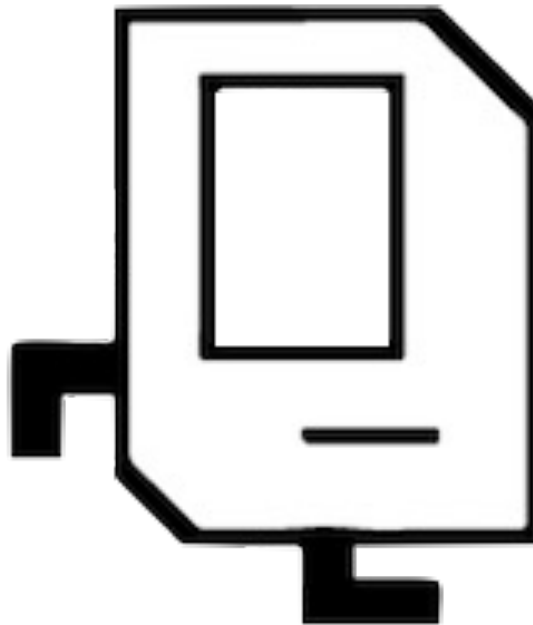


Let's learn!



Meet Karel the Robot

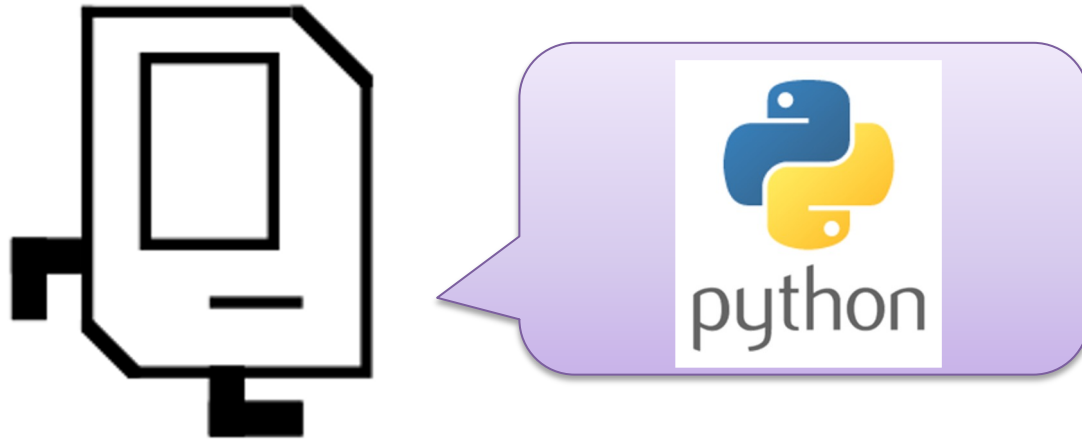
Rich Pattis



Good morning

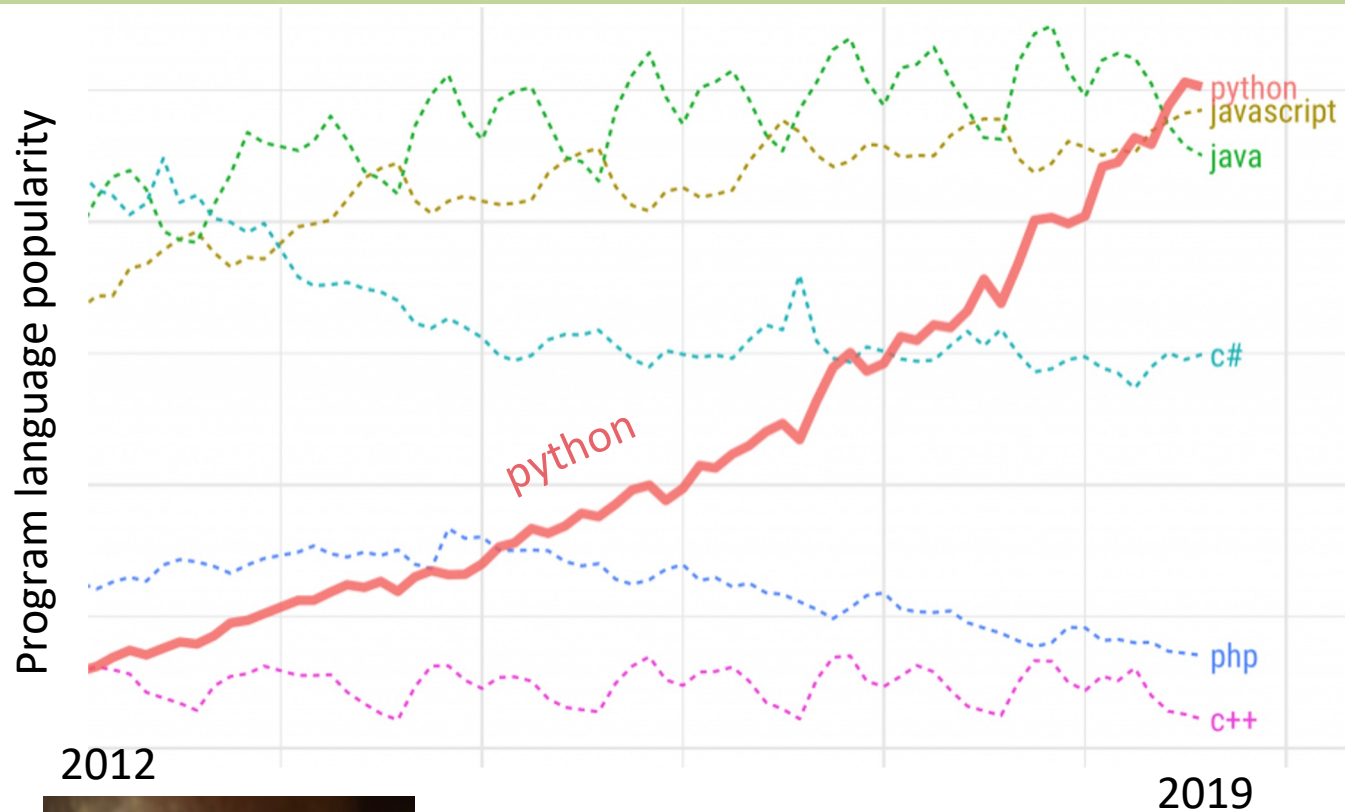


Karel Speaks Python



Why Python?

1



2

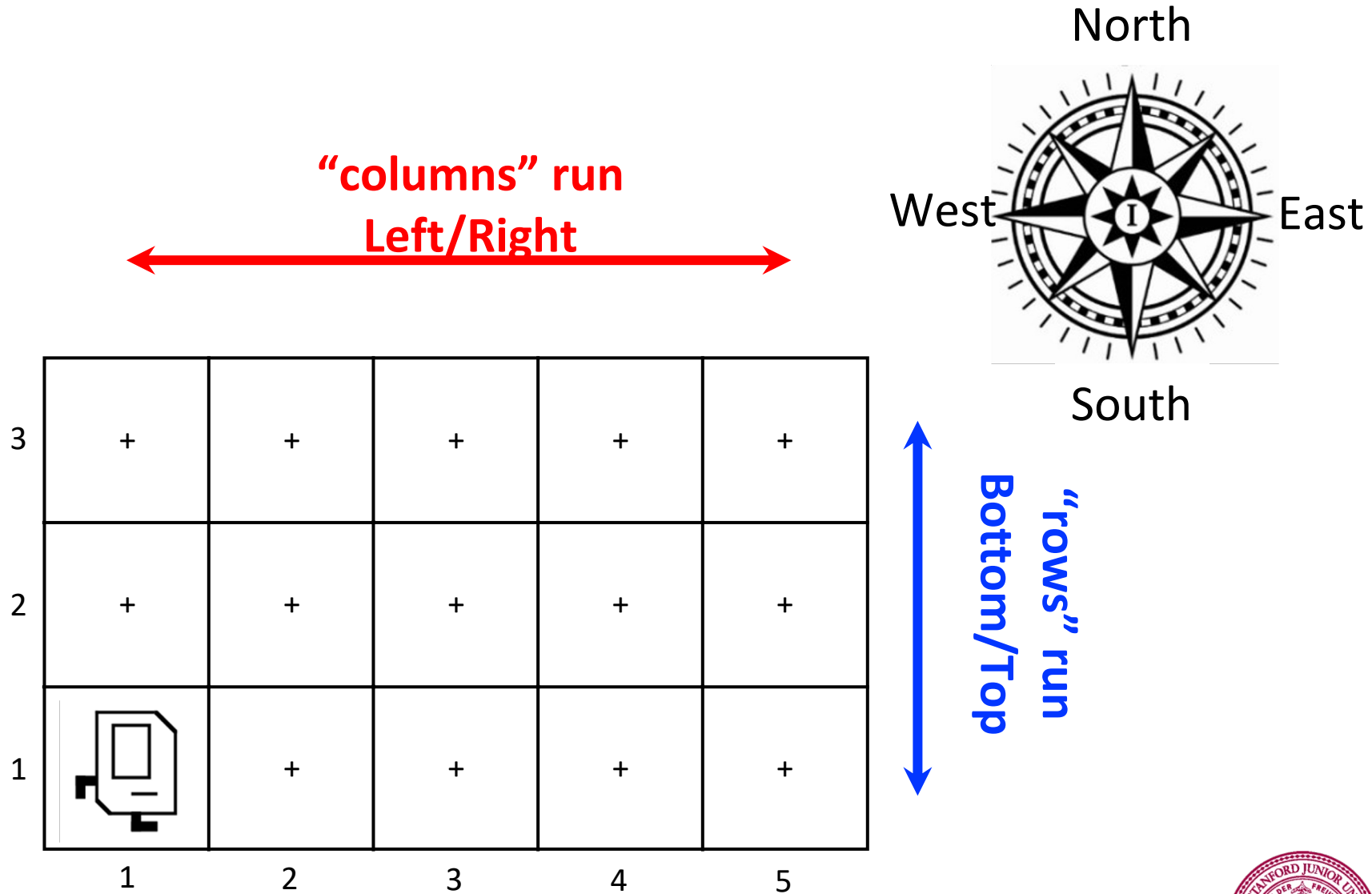


<https://stackoverflow.blog/2017/09/06/incredible-growth-python/>

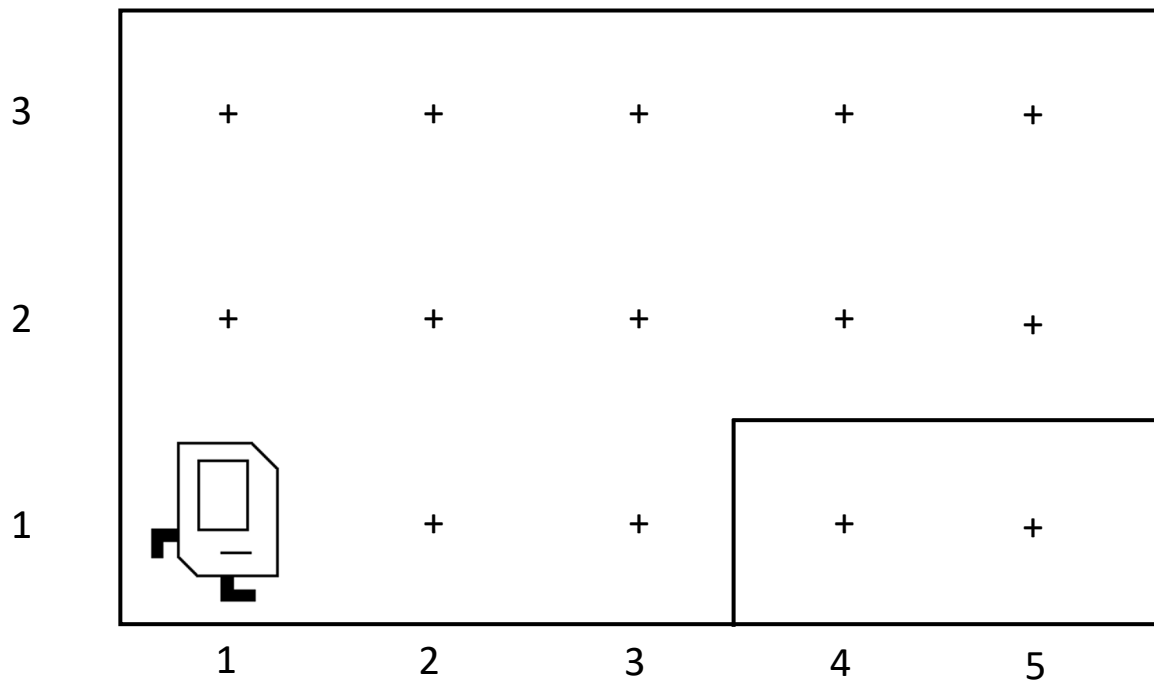
Guido van Rossum



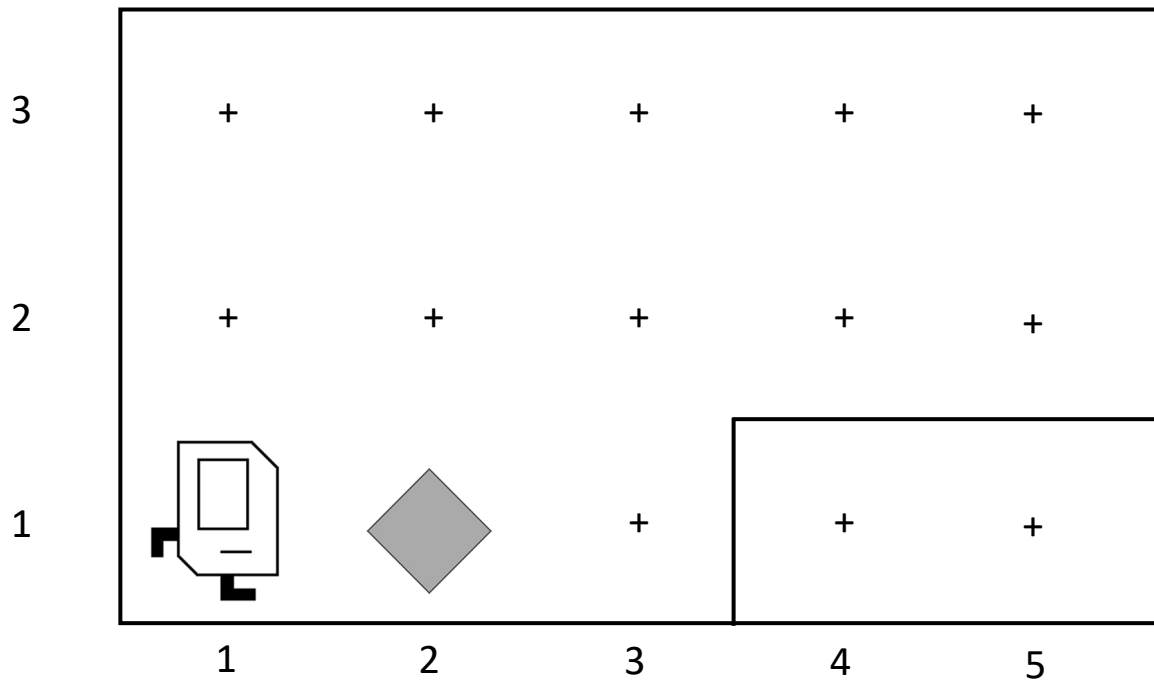
Karel's World



Walls



Beeper's



Knows Four Commands



`move()`

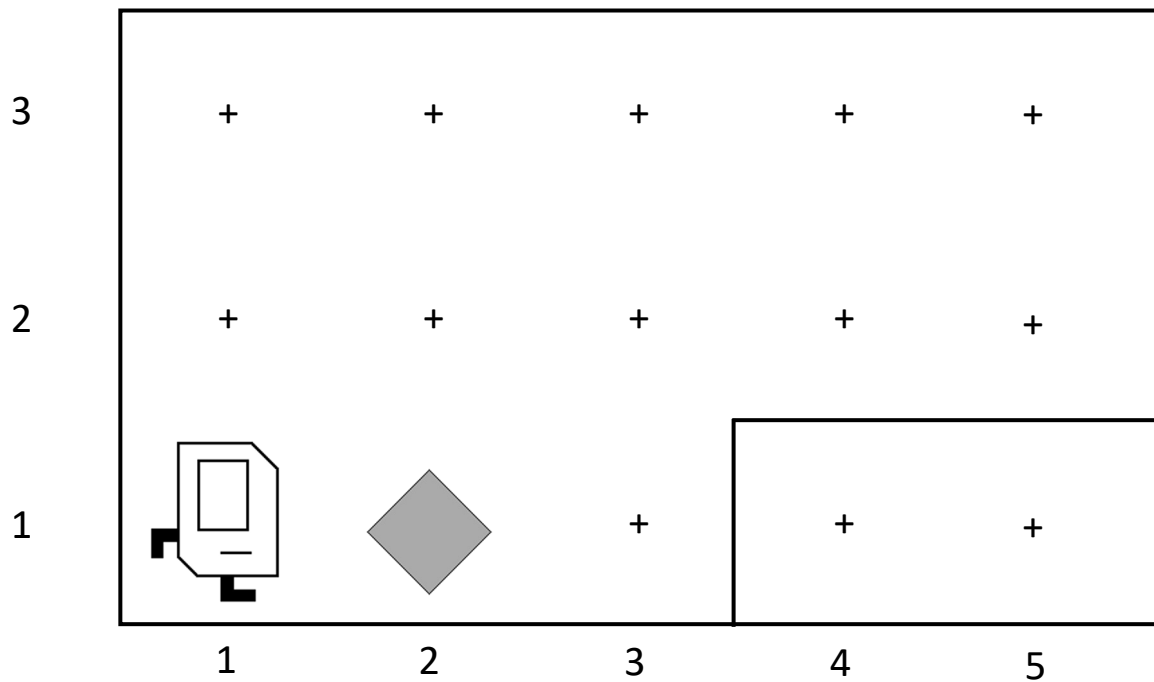
`turn_left()`

`put_beeper()`

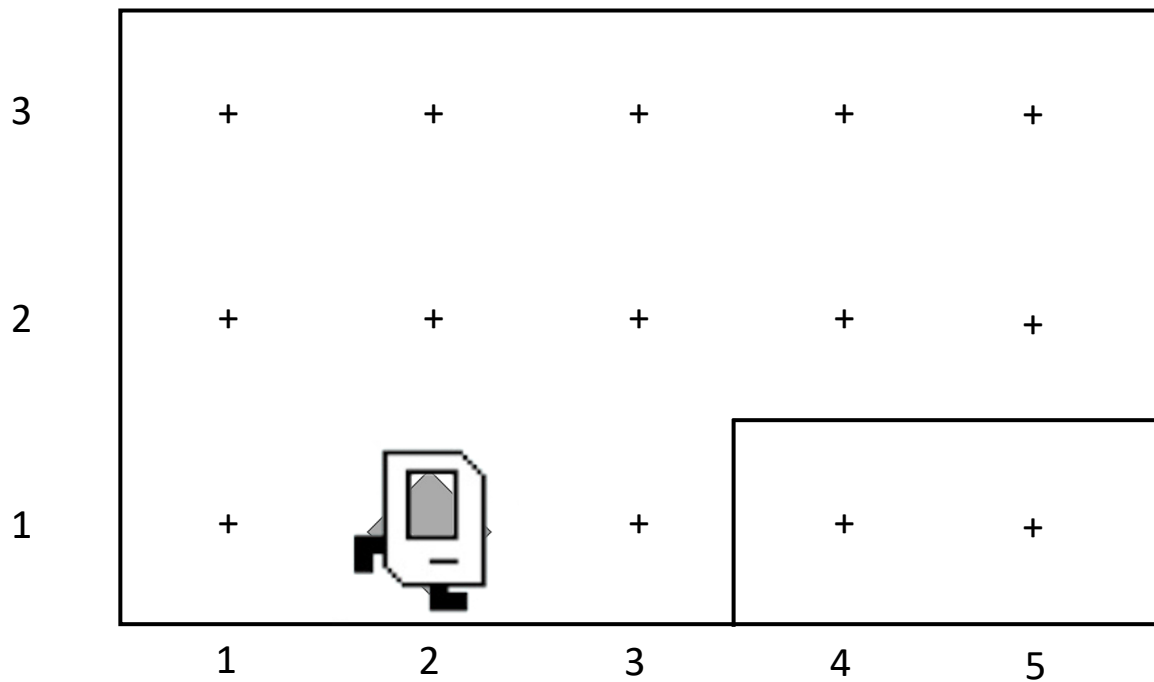
`pick_beeper()`

move ()

move ()

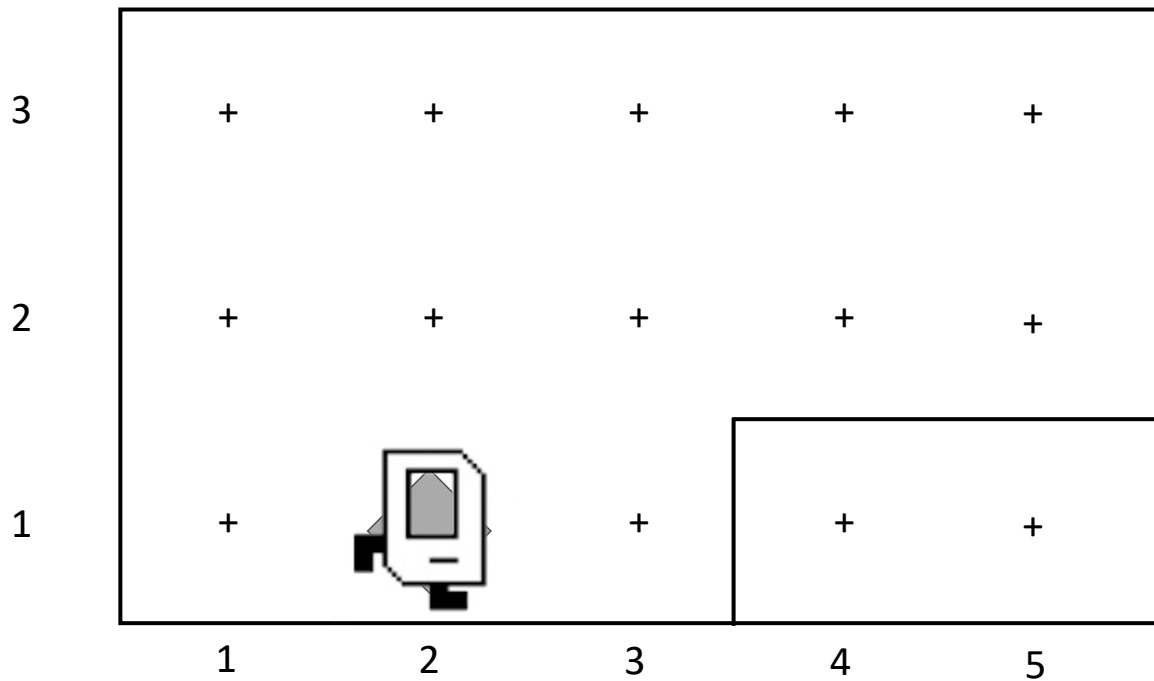


move ()

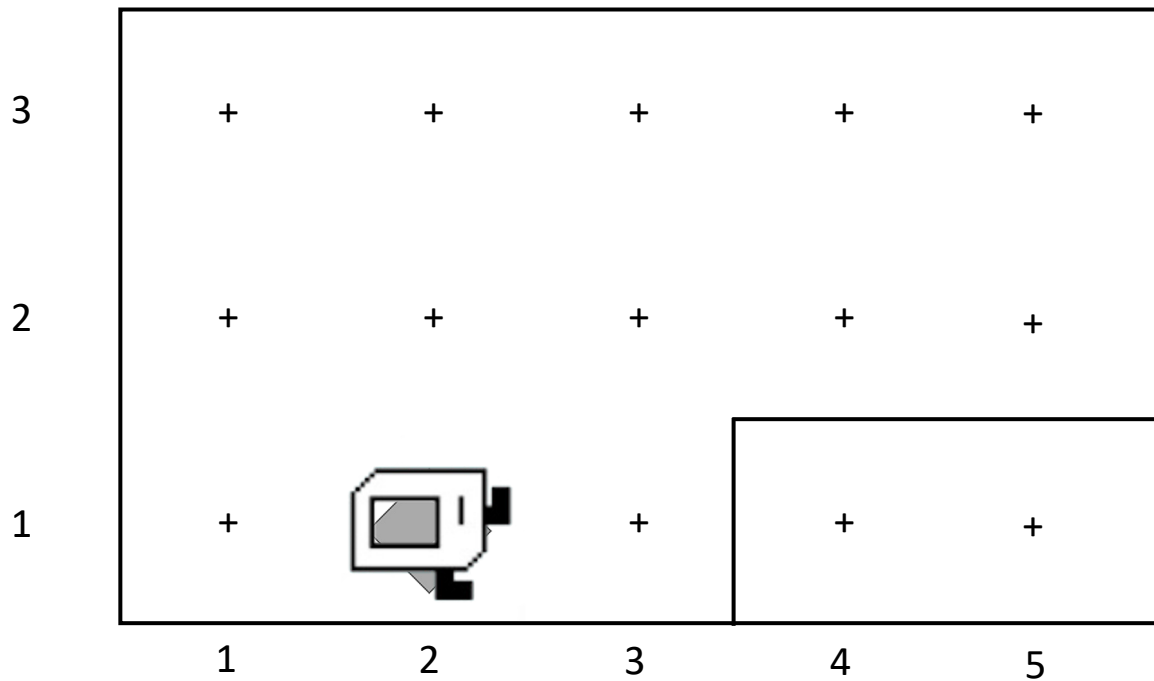


```
turn_left()
```

turn_left()

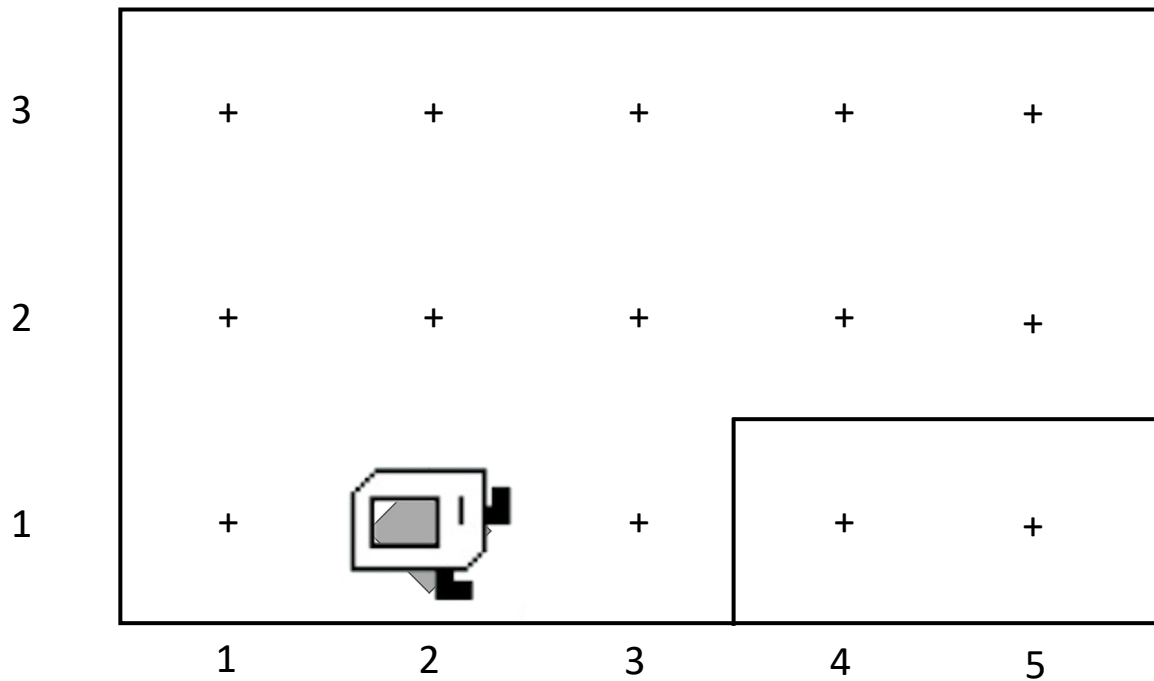


turn_left()

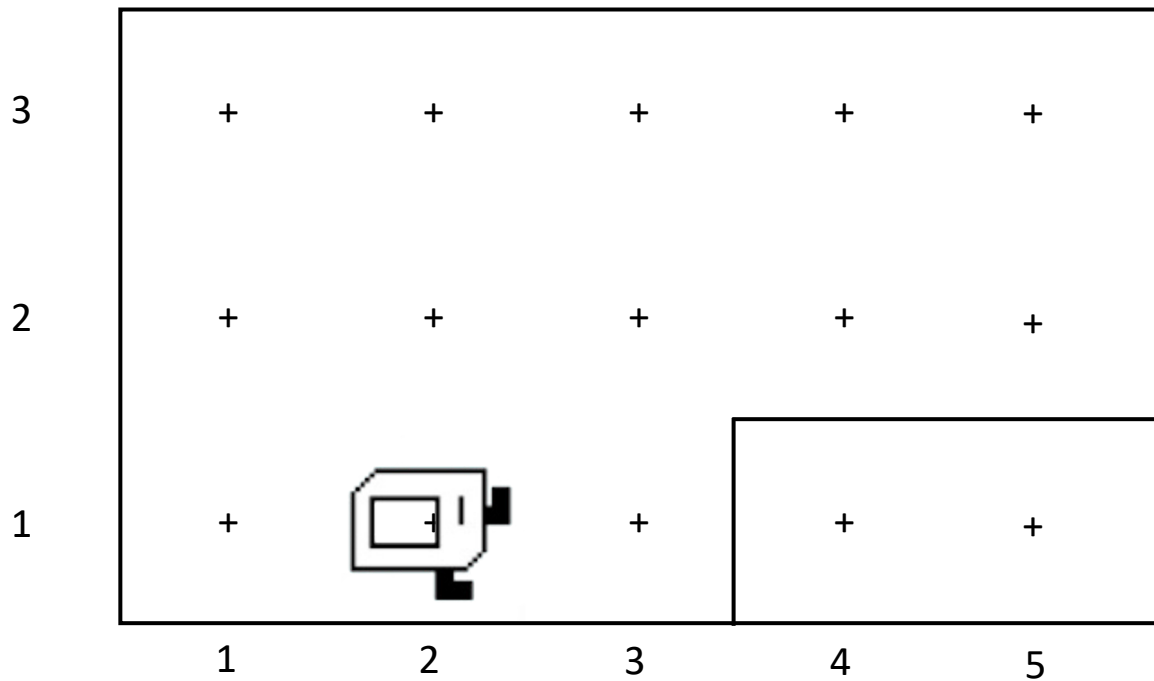



```
pick_beeper()
```

turn_left()

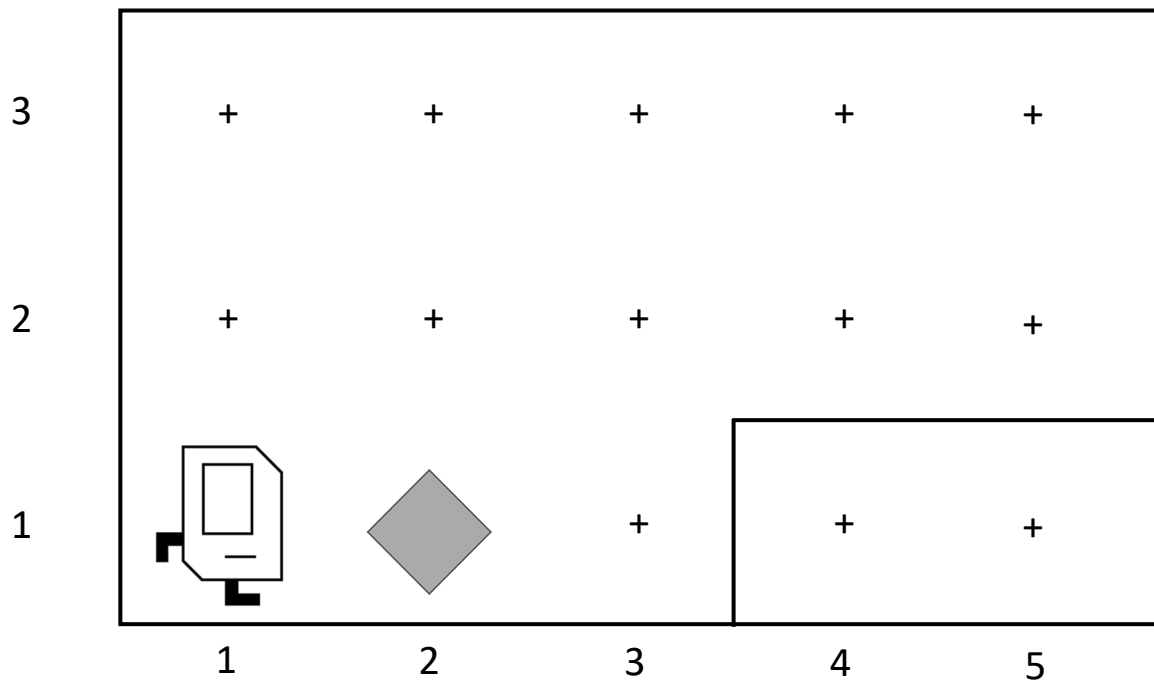


turn_left()

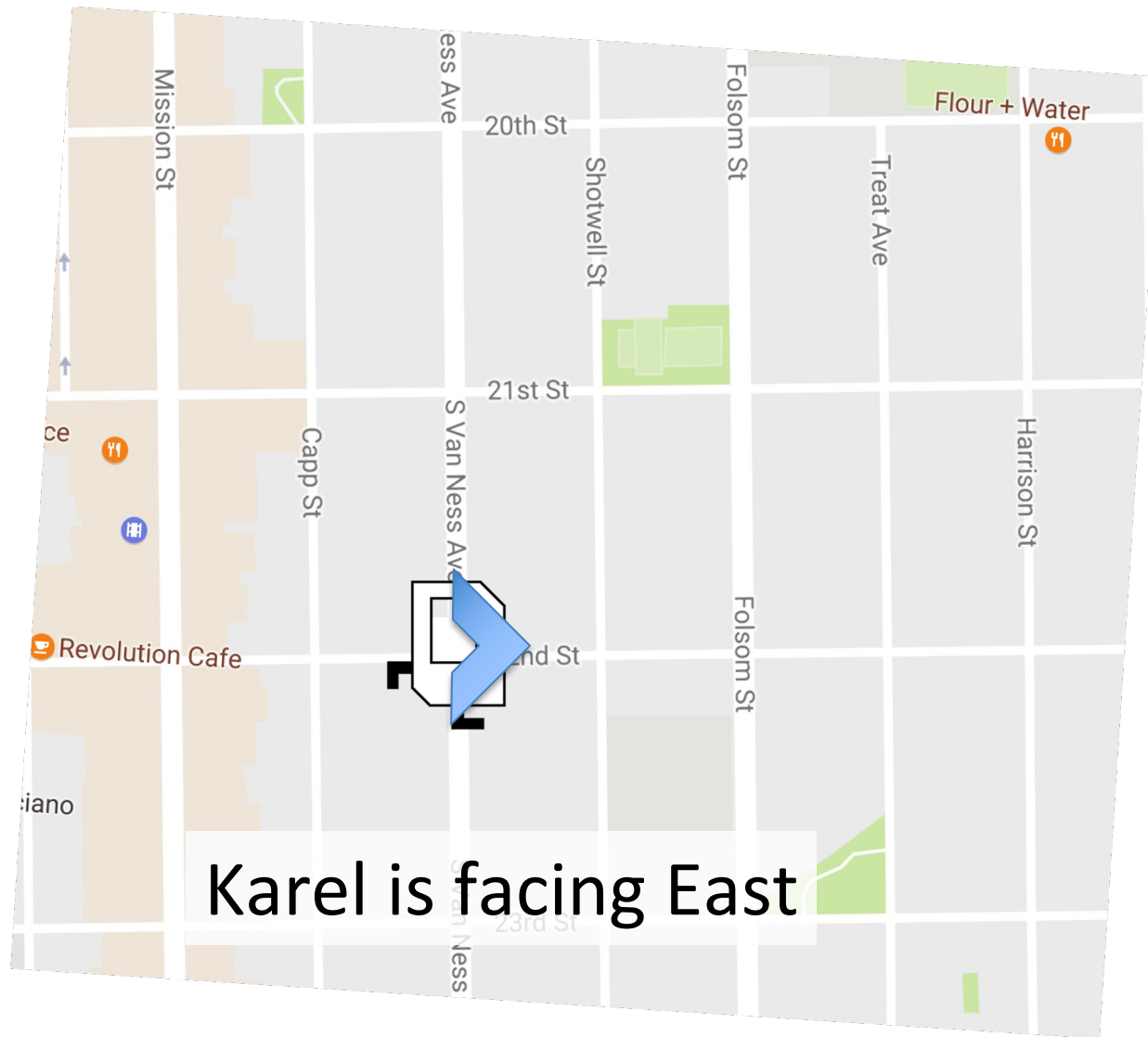


Make Sense?

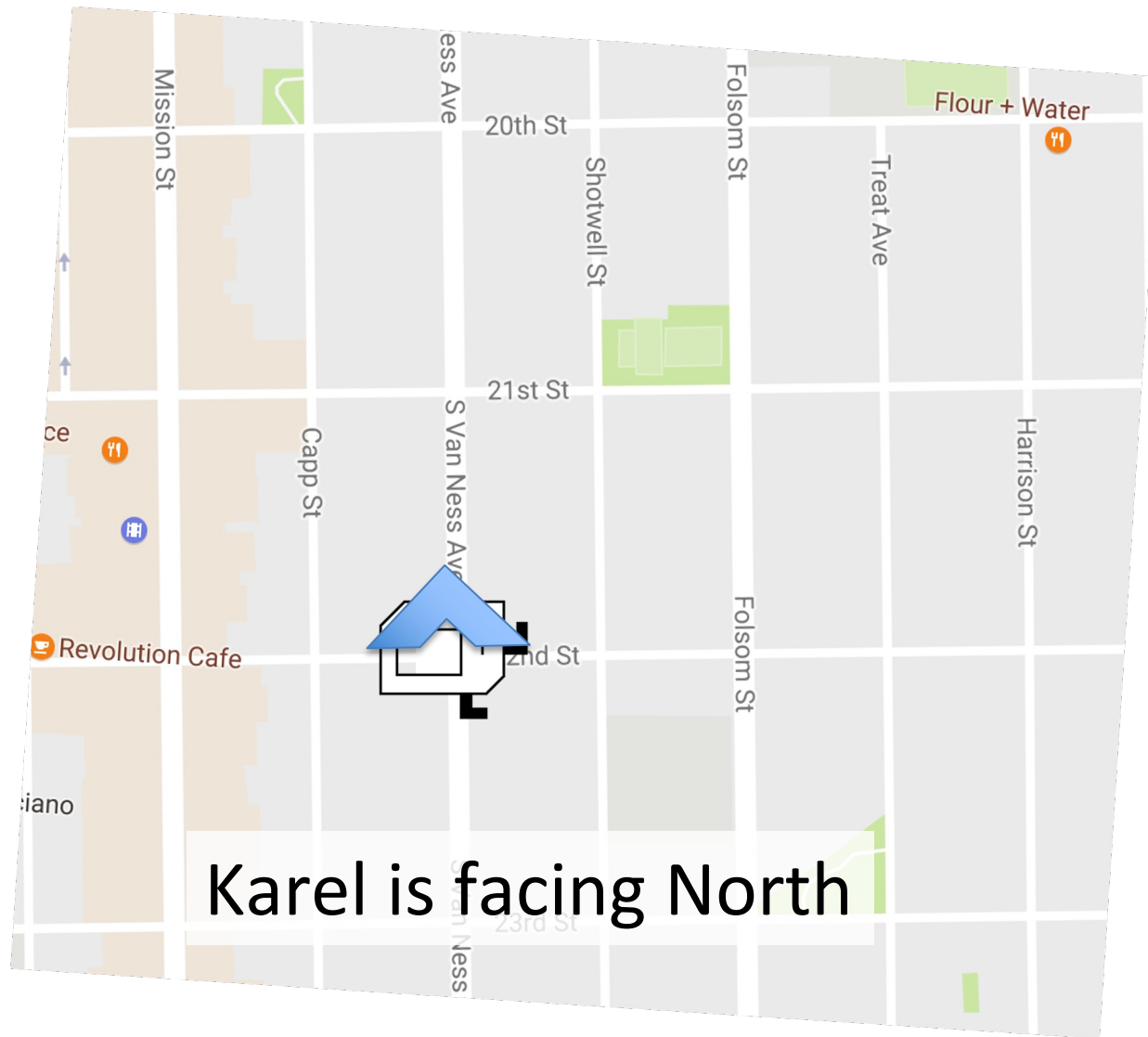
Bird's Eye View



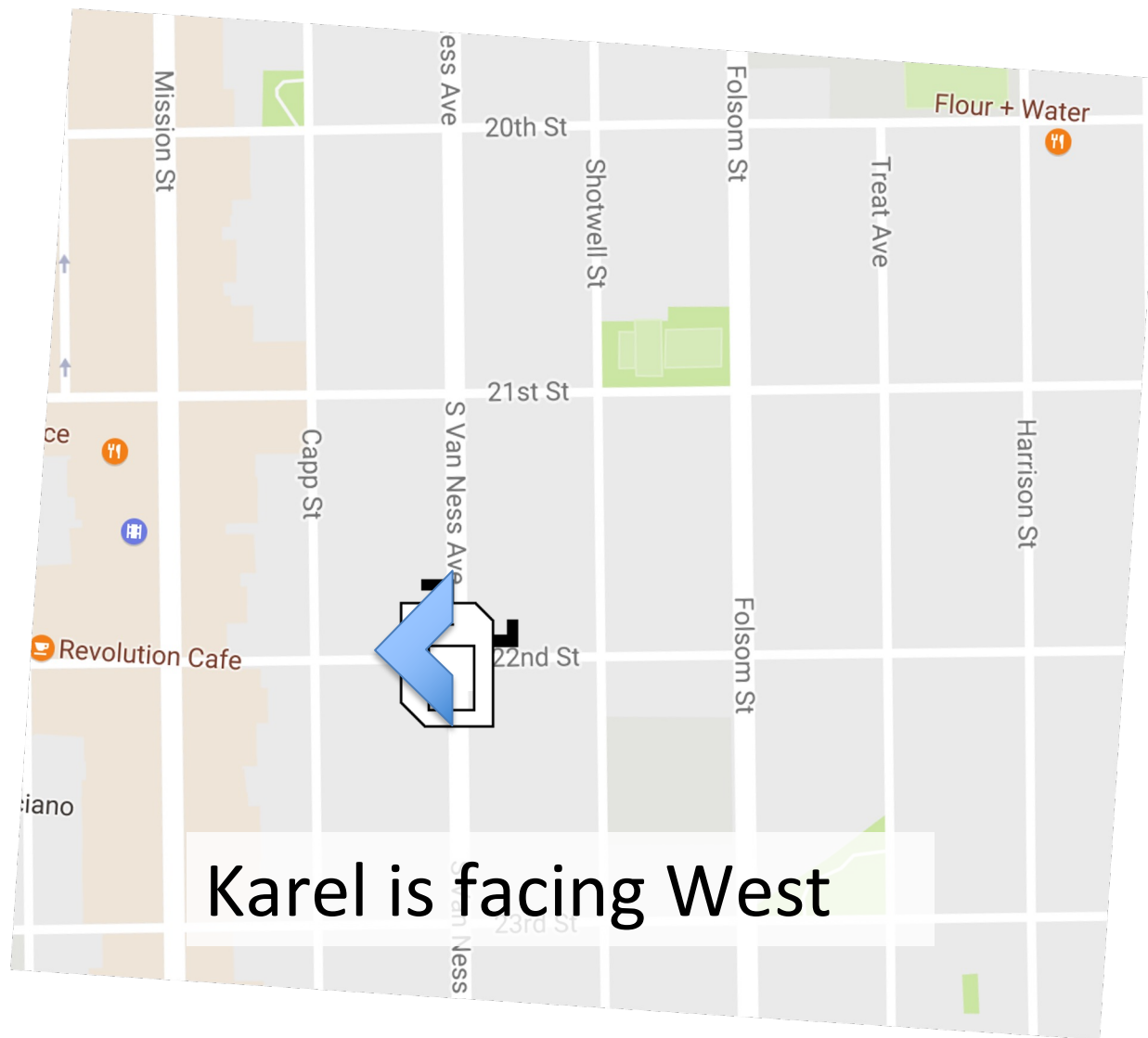
Bird's Eye View



Turn Left



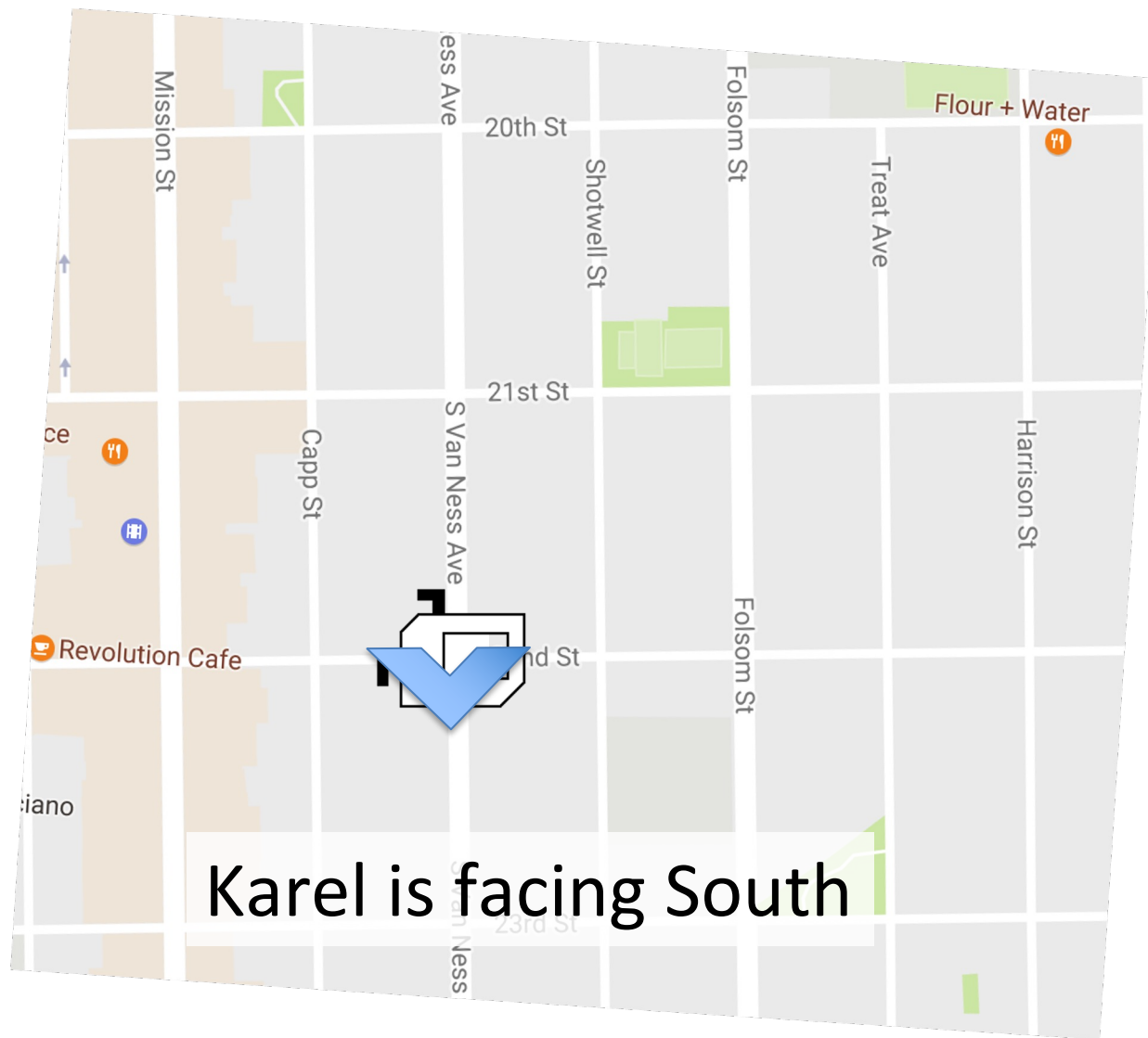
Turn Left



Karel is facing West



Turn Left

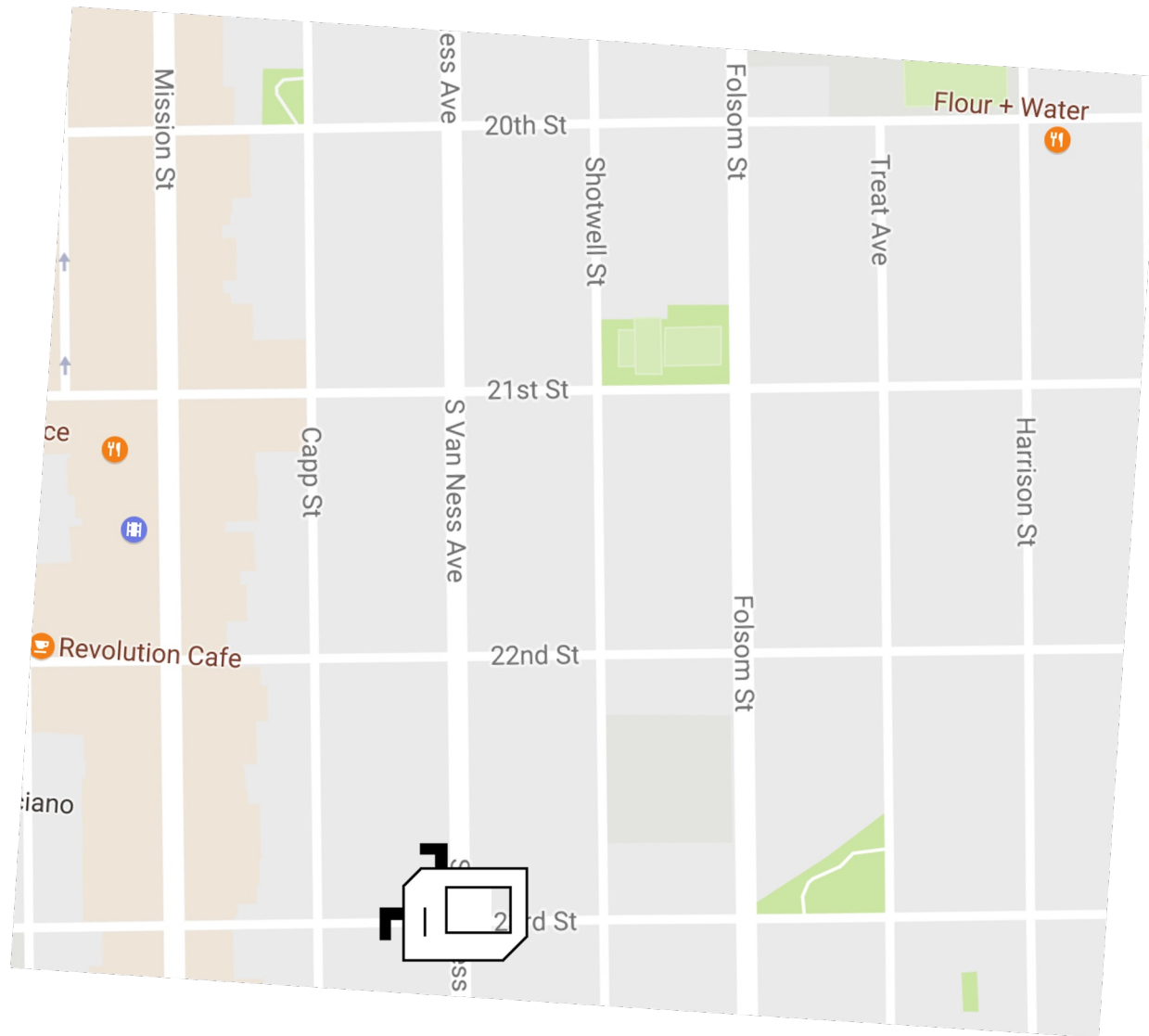


Karel is facing South

Piech and Sahami, Code in Place



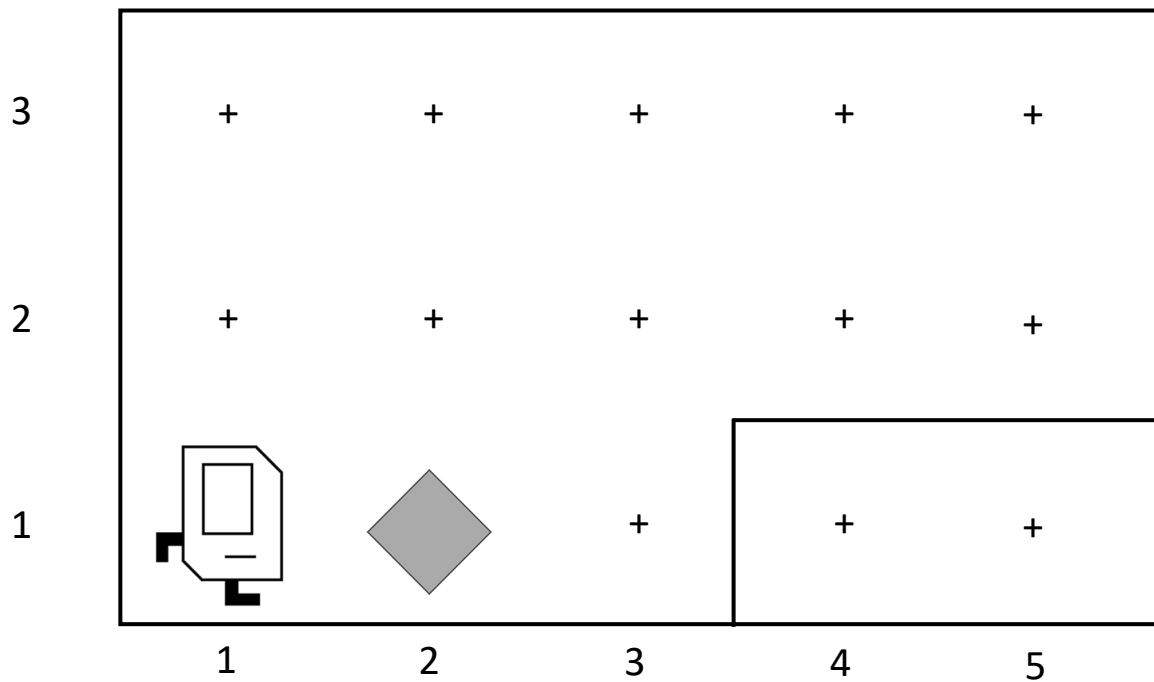
Move



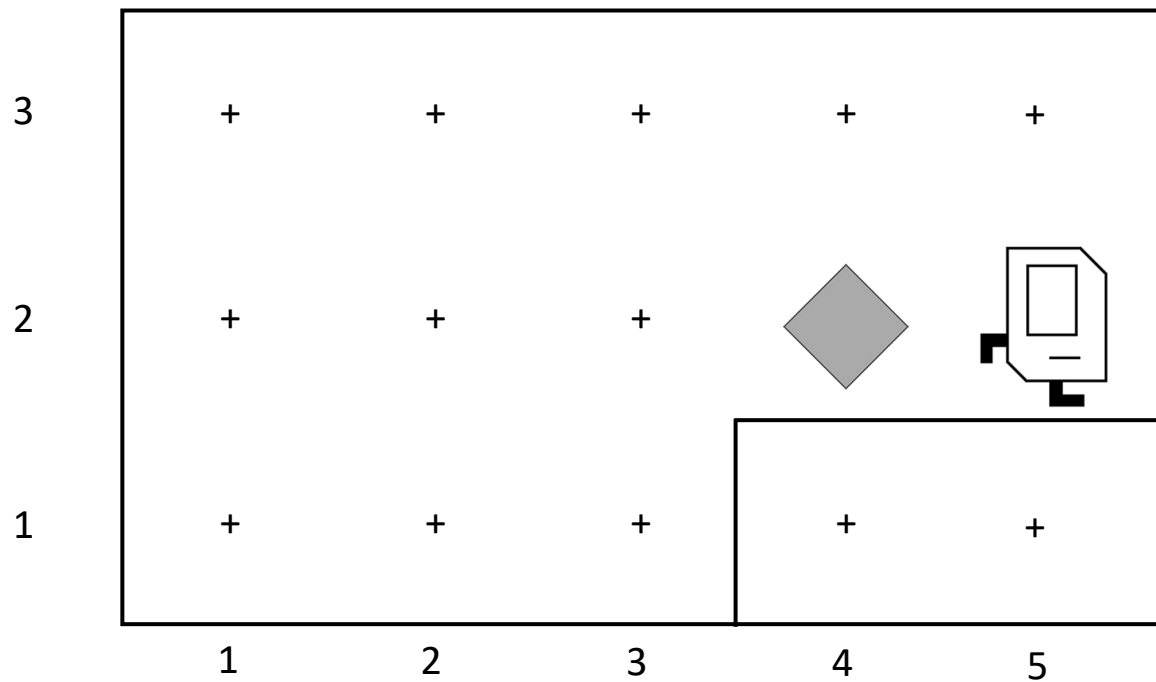
Piech and Sahami, Code in Place



First Challenge



First Challenge





Midjourney AI

Learn By Doing





The Python IDE
for Professional
Developers

DOWNLOAD

Full-fledged Professional or Free Community



Function Definition

```
def name() :  
    function statements
```

This adds a new
command to Karel's
vocabulary



Anatomy of a Program

Import Packages

Program



Anatomy of a Program

Import Packages



Anatomy of a Program

Import Packages

main function

helper functions

start program



Anatomy of a Program

Import Packages

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

helper functions

start program



Anatomy of a Program

Import Packages

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()  
  
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

start program



Anatomy of a Program

Import Packages

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()  
  
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()  
  
if __name__ == "__main__":  
    run_karel_program()
```



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

```
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

```
if __name__ == "__main__":  
    run_karel_program()
```



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

```
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

```
if __name__ == "__main__":  
    run_karel_program()
```



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

This piece of the program's **source code** is called a **function**.

```
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

```
if __name__ == "__main__":  
    run_karel_program()
```



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

This line of code gives the ***name*** of the function (here, the name is: **main**)

```
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

```
if __name__ == "__main__":  
    run_karel_program()
```



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

This line of code gives the *name* of the function
(here, the name is: **turn_right**)

```
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

```
if __name__ == "__main__":  
    run_karel_program()
```



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():
```

```
    move()
    pick_beeper()
    move()
    turn_left()
    move()
    turn_right()
    move()
    put_beeper()
    move()
```

This is called a **code block**
(Note the indenting)

```
def turn_right():
```

```
    turn_left()
    turn_left()
    turn_left()
```

```
if __name__ == "__main__":
    run_karel_program()
```



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

This is called a **code block**
(Note the indenting)

```
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

```
if __name__ == "__main__":  
    run_karel_program()
```



Anatomy of a Program

```
from karel.stanfordkarel import *
```

```
def main():  
    move()  
    pick_beeper()  
    move()  
    turn_left()  
    move()  
    turn_right()  
    move()  
    put_beeper()  
    move()
```

This is called a **code block**
(Note the indenting)

```
def turn_right():  
    turn_left()  
    turn_left()  
    turn_left()
```

```
if __name__ == "__main__":  
    run_karel_program()
```

